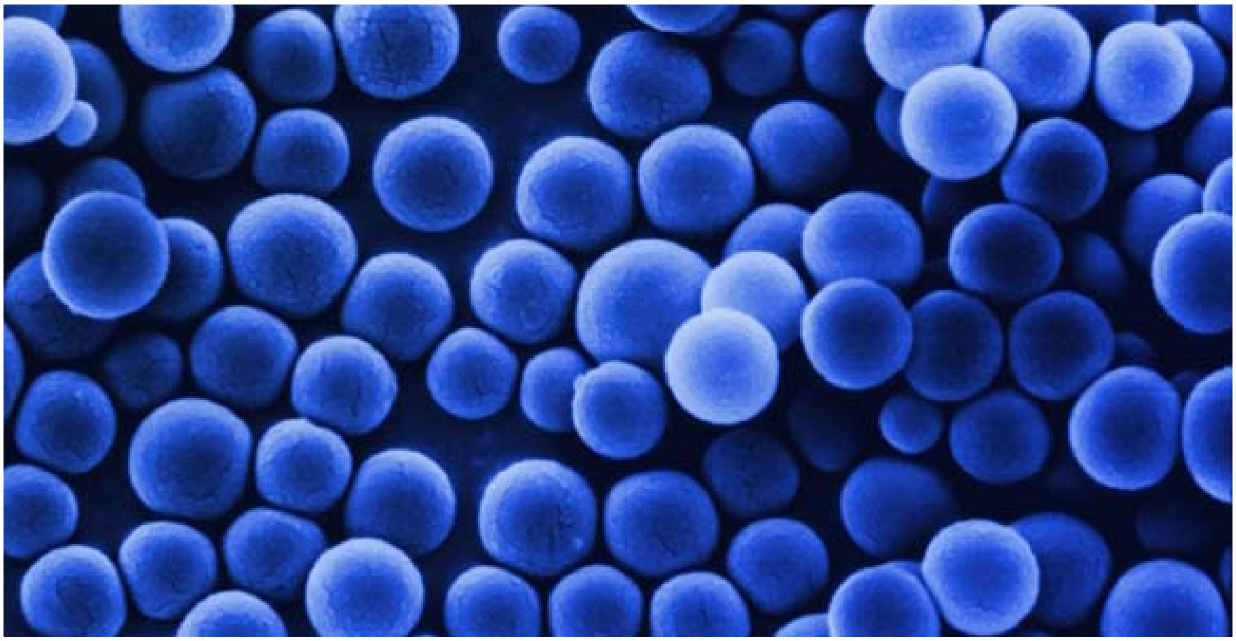
Seeding particles

For flow visualisation, LDA and PIV

Applications
For flow investigations by means of

- Flow Visualisation,
- Laser Doppler Anemometry and
- Particle Image Velocimetry

Features
- Optimum seeding for a wide range of applications
- Fluorescent particles give improved vector maps from PIV applications close to walls or through dirty windows
- Fluorescent particles enable simultaneous two-phase measurements with PIV (liquid-gas, liquid-solid)
- Non-toxic materials



Introduction
The application of optical flow measurement techniques to gas and liquid flows usually requires artificial seeding of the flow. Depending on the optical configuration and the overall conditions, tracer particles must fulfill different requirements. Results can be considerably improved by careful selection of the seeding material.

Description
PSP: Polyamide Seeding Particles are produced by polymerisation processes and therefore have a round but not exactly spherical shape. These microporous particles are highly recommendable for water flow applications.

HGS, S-HGS: Hollow Glass Spheres and Silver Coated Hollow Glass Spheres are borosilicate glass particles with a spherical shape and a smooth surface, preferrably for liquid flow applications. A thin silver coating further increases the reflectivity.

FPP: Fluorescent Polymer Particles are based on melamine resin. The fluorescence dye is homogeneously distributed over the entire particle volume. In applications with a high background light level, fluorescent seeding particles can improve the quality of vector maps from PIV and LDA measurements significantly. The receiving optics must be equipped with a filter centered around the emission wavelength.

DANTEC DYNAMICS

Ordering information

PSP, Polyamide Seeding Particles	
80A3011	PSP-5, diam. 5 µm, 250 g
80A4011	PSP-20, diam. 20 µm, 250 g
80A5011	PSP-50, diam. 50 µm, 250 g
HGS, Hollow Glass Spheres	
80A6011	HGS-10, mean diam. 10 µm, 250 g
S-HGS, Silver Coated Hollow Glass Spheres	
80A7001	S-HGS-10, mean diam. 10 µm, 100 g
FPP, Fluorescent Polymer Particles Excitation max.: 550 nm Emission max.: 590 nm	
80A8301	PMMA-RHB-10, 1-20um, 200 ml water suspension (50 g particles)
80A8311	PMMA-RHB-10, 1-20um, 400 ml water suspension (100 g particles)
80A8401	PMMA-RHB-35, 20-50um, 200 ml water suspension (50 g particles)
80A8411	PMMA-RHB-35, 20-50um, 400 ml water suspension (100 g particles)

Specifications

	PSP Polymid Seeding Particles	**HGS** Hollow Glass Spheres	**S-HGS** Silver Coated Hollow Glass Spheres	**FPP** Fluorecent Polymer Particles
Mean diameter (µm)				
Size distribution	1-10 µm 5-35 µm 30-70 µm	2-20 µm	2-20 µm	1-20 µm 20-40 µm 50-100 µm
Particle shape	Non-spherical but round	spherical	spherical	spherical
Density (g/cm^3)	1.03	1.1	1.4	1.5
Melting point (°C)	175	740	740	250
Refractive index	1.5	1.52	-	1.68
Material	Polymide 12	Borosilicate glass	Borosilicate glass	Melamine resin based polymer

Selection of seeding particles

	LDA	**Flow visualisation, PIV**
Gas flows	SAFEX fog generator *	Liquid Seeding Generator* High Volume Liquid Seeding Generator* Powder Seeder (for combustion experiments)*
Liquid flows	PSP – Polyamide Seeding Particles	PSP – Polyamide Seeding Particles HGS – Hollow Glass Spheres S-HGS – Silver Coated Hollow Glass Spheres FPP – Fluorescent Polymer Particles

* for details see separate product information.

The specifications in this document are subject to change without notice.